Volume 48 Number 3, Pt. I

BULLETIN
of the
AMERICAN ASSOCIATION OF
PETROLEUM GEOLOGISTS

MARCH, 1964

PALEOMAGNETIC CORRELATION OF UNITS WITHIN CHUGWATER (TRIASSIC) FORMATION, WEST-CENTRAL WYOMING[1]

M. DANE PICARD[2]
Lincoln, Nebraska

ABSTRACT

Directions of magnetization in the Triassic Chugwater Formation can be divided into two distinct groups nearly opposed to each other. These "normal" and "reversed" groups of magnetic directions have been used to correlate units within the Chugwater Formation in west-central Wyoming. Twenty-four possible magnetic "events" were noted; twelve of these appear to be caused by a reversal of the earth's magnetic field. Insufficient data are available to determine whether the other magnetic events record a reversal of the earth's magnetic field.

Local correlation between two sections 4.5 miles apart (northeast Thermopolis and south Thermopolis) indicates that the paleomagnetic correlation is not as precise as the lithologic correlation, but the two types of correlation do not conflict. Both types of correlation lines are approximately local "time-lines."

Regional correlation between stratigraphic sections along a cross section of 216 miles indicates that the paleomagnetic correlation lines are approximately "time-lines." Other results of the paleomagnetic correlation are: (1) lateral or vertical changes in rock type do not affect the magnetic directions within some intervals; (2) magnetic events differ in duration; (3) the number of magnetic events that can be correlated decreases as the distance between sections increases; and (4) the intensity of magnetization appears to be the same in normally and reversely magnetized samples.

The intensity of magnetization in the samples from the Chugwater Formation generally varies according to rock type. From least intense to most intense, this variation is: (1) drab siltstone and sandstone; (2) red, very fine-grained sandstone; and (3) red siltstone and poorly sorted siltstone. No significant relation between intensity of magnetization and bedding type has been noted.

INTRODUCTION

This paper describes the use of remanent magnetism to correlate stratigraphic units within the Chugwater Formation of west-central Wyoming. Directions of magnetization in these rocks are in two distinct groups nearly opposed to each other (Collinson and Runcorn, 1960; Black, 1960), implying that either the earth's magnetic field

[1] Manuscript received, March 26, 1963; revised, November 8, 1963.

[2] University of Nebraska.

Field work during the summers of 1961 and 1962, and two trips to California for paleomagnetic measurements, were supported by a National Science Foundation grant (G-14178) to F. B. Van Houten.

J. Vreeland assisted the writer during the first half of the 1961 field season. L. R. High, Jr., assisted the writer during the second half of the 1961 field season, all of the 1962 field season, and in the laboratory work.

R. R. Doell, Allan Cox, and E. Roth of the United States Geological Survey, Menlo Park, California, helped the writer with preliminary measurements. The Shell Oil Company permitted the use of a magnetometer at Los Angeles, California; the study would be uncompleted without this help. J. I. Gates and R. Bilohlavek were particularly helpful; the hospitality of Don Hughes is acknowledged.

Allan Cox, R. R. Doell, F. B. Van Houten, A. F. Buddington, R. B. Hargraves, S. B. Lumbers, and L. D. Ayres read and criticized preliminary drafts of the manuscript. The drafting is by John Wells, and was supported by a National Science Foundation grant (GP-110). The writer's wife, Virginia, and Jean Pollock typed various drafts of the manuscript.

This paper is drawn from material submitted in partial fulfillment of the requirements for the degree of Doctor of Philosophy at Princeton University. Additional work was done subsequently.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

has periodically reversed itself, or the rocks have become magnetized in a direction opposite to that of the earth's field by a self-reversal process (Cox and Doell, 1960, p. 652–653). Reversals of the earth's magnetic field appear to be responsible in the Chugwater Formation, offering the possibility of correlating strata that became magnetized at the same time.

For the present study, oriented cores were collected from five measured sections of the Chugwater (Triassic) Formation in west-central Wyoming. The mean direction of magnetization for the two predominant magnetic directions is 343° declination (clockwise from 0°), +46° inclination (+downward) for one group termed *normal*, and 141½° declination, −31° inclination (−above horizontal) for a second group termed *reversed*. In a typical vertical sequence, numerous normal and reversed magnetic zones are superposed and these may be traced from locality to locality. At least twelve such normal or reversed zones occur within the Chugwater.

Sampling and Measurement

The term oriented "core" is used to describe an individually oriented piece of rock; "sample" is used to describe pieces of rock cut from an oriented core and used in the measuring apparatus.

Oriented cores were collected by drilling 1-inch-diameter cores with a portable water-cooled diamond drill. Cores were 2–4 inches in length, and their orientation was measured with a Brunton compass while still attached to the outcrop. Orientation errors rarely exceeded 3°. The cores were cut into samples 0.90 inch long. The remanent magnetism of one sample from each core, except for the duplicate samples noted in Table I, was measured with a spinner magnetometer. Eleven samples were measured at the United States Geological Survey laboratory in Menlo Park, California; 282 samples were measured at the Shell Oil Company's production laboratory in Los Angeles, California. Cores were drilled at diverse angles in the field, and were rotated to horizontality after measurement of the magnetization. Measurements of the same sample on both instruments indicate that measurements of directions of magnetization are accurate within 2°. Tests of consistency between two samples from the same core (Table I) indicate an average difference between the samples of 4.1° in declination, 3.2° in inclination, and 0.86 ($\times 10^{-6}$ emu/cc) in intensity. Forty-two per cent of the intersections of the three planes that were measured on the magnetometers intersect on a stereonet at an angle of 1° or less, 63 per cent at an angle of 2° or less, and 85 per cent at an angle of 4° or less.

General Stratigraphy and Nomenclature of Chugwater Formation

The nomenclature of the Chugwater Formation proposed by Love (1939) and followed by McKee et al. (1959) is used here with modifications. From oldest to youngest, Love's units of the Chugwater Formation are: Red Peak Member; Alcova Limestone Member; Crow Mountain Sandstone Member; and Popo Agie Member. The writer subdivides the Red Peak Member into five informal rock units; from oldest to youngest these are: silty claystone facies, lower platy facies, alternating facies, upper platy facies, and variegated zone. These subdivisions have been recognized over much of the Bighorn and Wind River basins

Table I. Comparison of Different Samples from Same Core
(Declination from True North: Inclination from horizontal; +, lower hemisphere; −, upper hemisphere)

No.	*Decl.; Incl.; (degrees) Intensity ($\times 10^{-6}$ emu/cc)*		*Diff. in Decl.*	*Diff. in Incl.*	*Diff. in Intensity ($\times 10^{-6}$)*	*Rock Type*
R-8	77; 66−; 34.8	72; 64−; 31.8	5°	2°	3.0	sltst.
R-39	253; 20−; 6.5	254; 18−; 6.5	1°	2°		p.s. sltst.
R-56	10; 24+; 0.7	10; 26+; 0.7		2°		v.f. ss.
C-37	126; 32−; 7.2	129; 31−; 7.4	3°	1°	0.2	sltst.
T-28	202; 17−; 5.2	189; 17−; 4.9	13°		0.3	sltst.
T-37	63; 27−; 7.1	61; 32−; 7.0	2°	5°	0.1	sltst.
T-40	73; 55−; 6.8	67; 52−; 4.8	6°	3°	2.0	p.s. sltst.
T-42	68; 33−; 7.2	65; 22−; 5.9	3°	11°	1.3	sltst.
	Average difference		4.1°	3.2°	0.86	

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

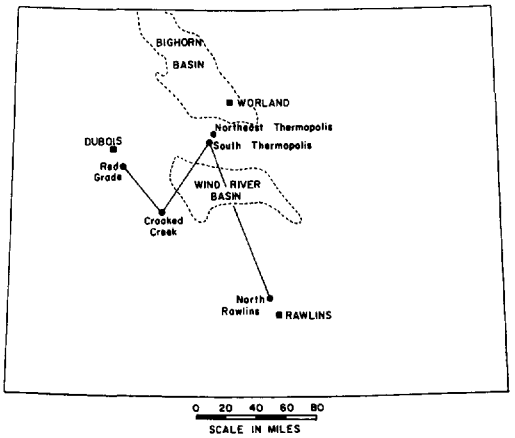


FIG. 1.—Index map. Locations of measured sections, west-central Wyoming.

(Fig. 1); they are characterized by variations in kind and amount of particular rock types, and by bedding characteristics. The alternating facies has been described briefly (Picard and High, 1963).

The maximum dip of the Chugwater Formation in the sections sampled is 12°, commonly in a northwest direction. The line of section through the sample localities (Fig. 1) is oblique to the former position of the miogeosyncline of early Triassic age (Kummel, 1954). The precise location of each sample locality is given in Picard (1963).

ROCK TYPES AND COLORS

The terminology for the fine-grained rocks described here is from a previous paper (Picard and High, 1963, p. 89). The rock type of each sample is shown on Figure 2, and, where appropriate, in tables throughout the paper. The percentage of each rock type among all samples collected is: sandstone, 22.3 per cent; siltstone, 61.9 per cent; poorly sorted siltstone, 12.4 per cent; and silty claystone, 3.4 per cent.

Rock color was determined on about 100 wet samples by comparison with the National Research Council color chart (Goddard et al., 1948). The most common color is grayish red (about 10 R 4/2); pale reddish brown (about 10 R 5/4) is next in abundance.

MINERALOGY

General.—Numerous samples of whole rocks with normal and reversed magnetic directions were examined to see if there is any difference detectable by X-ray diffractometer in the nonopaque mineralogy of normal and reversed samples. No consistent differences were noted. The mineralogical data acquired in this study that is relevant to the magnetic properties of the rocks is as follows.

X-ray analyses of opaque minerals.—Iron oxides can not be generally detected on any of the X-ray diffraction patterns of whole rock samples, but hematite is probably present on a few of the X-ray patterns.

Hematite, ilmenite, and magnetite were identified by X-ray diffractometer in several opaque mineral separations from red siltstone and very fine-grained sandstone. In these separations, hematite is predominant, and ilmenite appears to be more common than magnetite. X-ray patterns of standard mixtures of hematite, ilmenite, and magnetite were compared with the X-ray patterns of the Chugwater Formation. From X-ray analyses, Van Houten (1961, p. 297) indicated that specular hematite is the dominant iron oxide in the Chugwater Formation.

Hematite and goethite(?) were identified by X-ray diffractometer in an opaque mineral separation from a drab, very fine-grained sandstone.

Further observations on opaque minerals.—Modal determinations of thirteen thin sections from rocks of the Red Peak and Crow Mountain Sandstone members indicate a range in opaque minerals from about 0.7 to 3.7 per cent (Table II). The opaque minerals in the siltstones and poorly sorted siltstones have an average diameter of about 0.05 mm., ranging from 0.005 to 0.153 mm. The opaque minerals in the sandstones average about 0.06 mm. in diameter. Although there are but few data, these suggest (Table II) that red siltstone may generally contain a higher per cent of opaque minerals than red, poorly sorted siltstone.

No polished sections were studied, but on the basis of examination of opaque mineral separations with the binocular microscope, and examination of thin sections in reflected light, the following additional observations are tentatively presented. These observations were made subsequent to the more general modal determinations.[3]

Hematite is the dominant opaque mineral, occurring in localized patches, in small veinlets

[3] It may be noted that the writer believes that he can distinguish magnetite from hematite in thin sections in reflected light in these rocks.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

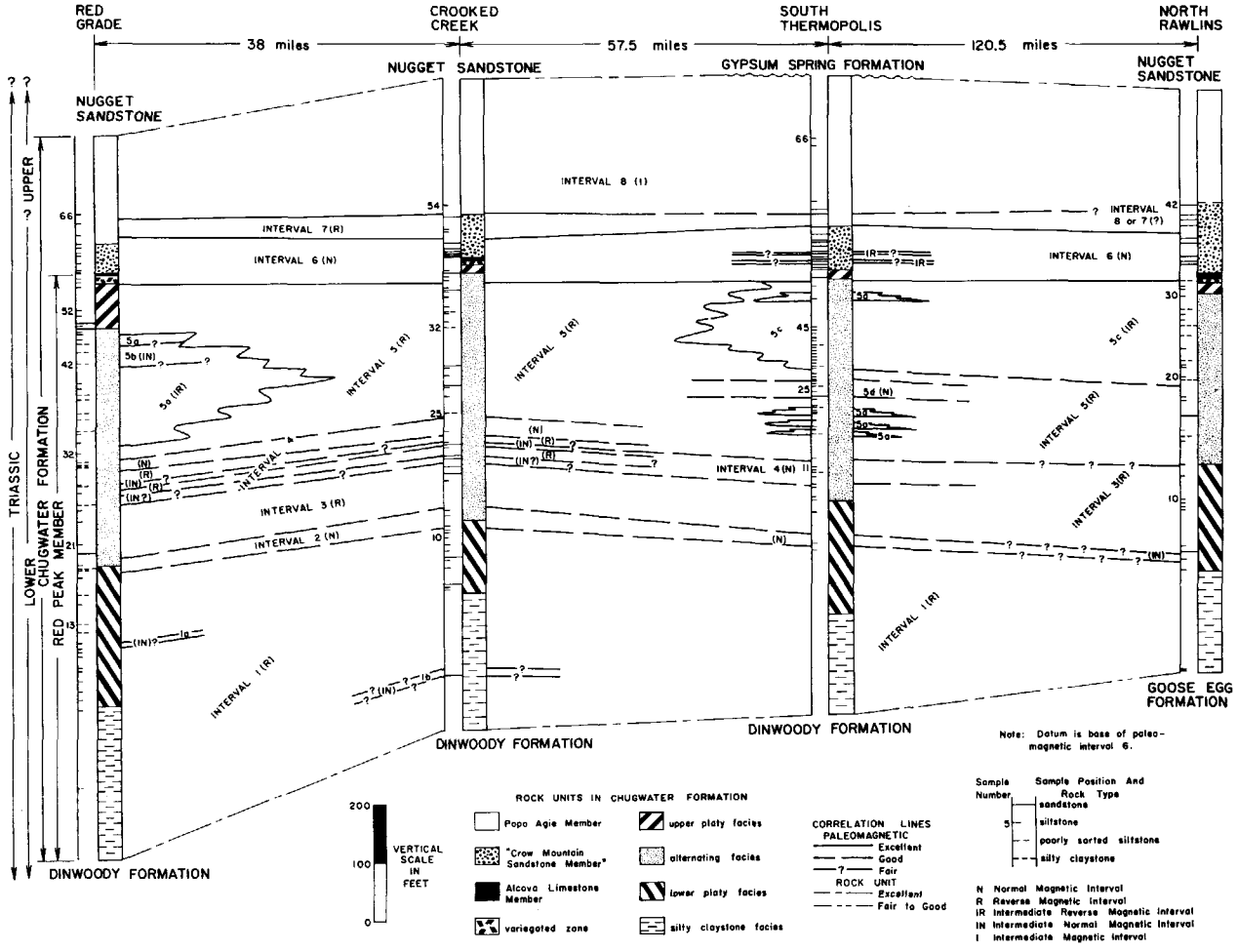


FIG. 2.—Regional cross section showing paleomagnetic correlation between Red Grade and north Rawlins sections.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin University Library user on 15 June 2026

TABLE II. MODAL DETERMINATIONS OF OPAQUE MINERALS AND RED PIGMENT
(2,000 points counted per thin section)

No.	*% Opaque Minerals*	*% Red Pigment*	*% Total Iron*	*Intensity* ($\times 10^{-6}$ *emu/cc*)	*Magnetic Direction*	*Rock Type*
T-15	0.7	16.7	2.16	5.9	IR	p.s. sltst.
T-9	0.9	7.6	1.94	6.6	N	p.s. sltst.
T-58	1.0	2.7		1.1	N	v.f. ss.
TN-1b	1.3	1.1		0.6		drab sltst.
NR-14	1.5	9.6	2.13	4.7	R	p.s. sltst.
C-47	1.7	6.4		4.8	N	v.f. ss.
C-35	1.7	7.3	1.98	7.3	R	p.s. sltst.
T-49	1.1	5.1	0.89	7.0	N	v.f. ss.
C-25	2.0	14.4	1.35	7.2	R	p.s. sltst.
C-40	2.1	1.4	1.13	6.5	R	sltst.
R-1b	2.3	9.9		11.1 (R-1)	R (R-1)	p.s. sltst.
NR-24	2.4	2.8	0.88	22.3	IR	sltst.
NR-20	3.7	3.7		61.8	IR	sltst.

and lenses, as pore space "filler," and as detrital grains. Many of the detrital grains of hematite are rounded, and in a few grains red staining radiates outward from hematite grains a fraction of a millimeter. Hematite is much more common than magnetite, and rounded hematite grains are more common than rounded magnetite grains.

Magnetite, hematite, ilmenite, leucoxene, and unidentified opaque minerals form concentrates in thin laminae in some rocks. The three samples with the highest percentage of opaque minerals (Table II) are characterized by concentrations of opaque minerals in thin laminae that parallel the bedding. Rounded, unaltered detrital magnetite also occurs as small grains disseminated in red pigment, and as disseminated detrital grains. Commonly, detrital magnetite is altered to hematite, with gradations from tiny spots of hematite to thin skins of hematite surrounding the magnetite grains. Hematite, derived from magnetite, radiates outward a fraction of a millimeter from grains of magnetite in some rocks.

Pyrite, or other sulphides, were not noted in any of the samples that were examined. Maghemite was also not noted, but a detailed search for its presence was not made.

Red pigment.—Red pigment, in the thirteen thin sections that were counted for opaque minerals, ranges from 1.4 to 16.7 per cent (Table II). It is difficult, however, to determine the amount of red pigment in a particular sample because of variations in the intensity of red pigmentation. In some rocks the red pigment is only a thin, superficial coating on the matrix material. Red pigment variously occurs as grain coating, and in pockets, lenses, stringers, and veinlets. The red pigment occupies pore space in the rocks, and is distributed irregularly throughout the rocks. It is unlikely that enough hematite as grain coating was brought into the depositional area to give the per cent hematite present.

PALEOMAGNETIC APPLICABILITY OF SAMPLES

General.—The paleomagneticist studying a group of rocks is primarily interested in the average direction of the magnetic field at the locality where the rocks were sampled, and the geologic date when the field had that direction. Several criteria can be used (Graham, 1949; Cox and Doell, 1960, p. 655–661) to judge the stability of samples. The following paragraphs discuss some of the evidence pertaining to the stability of samples from the Chugwater Formation.

Tests of stability.—Directions of magnetization that are strongly divergent from the present field direction indicate that there is no dominant component of magnetization parallel with the present field. Cox and Doell (1960, p. 656) state that, "Parallelism between tightly grouped mean directions of magnetization in two groups of samples which are reversely magnetized with respect to each other is a much stronger test than simple consistency of directions without reversals." The samples collected from the Chugwater Formation have magnetic directions that are consistently different from the present earth's field, and have two dominant magnetic directions that are reversed with respect to each other. The average magnetic directions of normal and reversed samples from the Chugwater Formation are consistently about 180° apart in declination, but about 160° apart in inclination. The fact

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

TABLE III. COMPARISON OF RESULTS RECORDED FROM SAME SAMPLE AFTER SIX-MONTH INTERVAL

No.	*Decl.; Incl.; Intensity (× 10⁻⁶ emu/cc)* *January, 1962*	*July, 1962*	*Diff. in Decl.*	*Diff. in Incl.*	*Change in Intensity (× 10⁻⁶)*	*Rock Type*
R-15	133; 2−; 6.4	134; 1−; 5.8	1°	1°	−0.6	p.s. sltst.
R-33	273; 30−; 16.3	273; 36−; 6.9		6°	−9.4	sltst.
R-39	253; 20−; 6.9	254; 18−; 6.5	1°	2°	−0.4	p.s. sltst.
R-41	234; 33−; 7.6	236; 33−; 7.1	2°		−0.5	sltst.
R-61	349; 51+; 5.7	352; 51+; 4.9	3°		−0.8	sltst.
R-63	45; 12+; 5.0	46; 11+; 2.8	1°	1°	−2.2	sltst.
C-35	145; 23−; 7.3	145; 18−; 6.6		5°	−0.7	p.s. sltst.
C-37	129; 31−; 7.4	135; 31−; 7.1	6°		−0.3	sltst.
C-42	5; 54+; 7.2	3; 56+; 6.6	2°	2°	−0.6	sltst.
C-48	332; 40+; 5.3	344; 45+; 2.5	12°	5°	−2.8	v.f. ss.
NR-17	155; 14+; 3.8	155; 15+; 4.0		1°	+0.2	sltst.
NR-27	80; 37−; 7.1	76; 41−; 6.5	4°	4°	−0.6	sltst.
NR-35	340; 27+; 7.2	338; 21+; 6.4	2°	6°	−0.8	v.f. ss.
T-40	73; 55−; 6.8	67; 52−; 4.8	6°	3°	−2.0	p.s. sltst.
T-41	78; 22−; 18.3	39; 27−; 6.0	39°	5°	−12.3	sltst.
T-42	68; 33−; 7.2	65; 22−; 5.9	3°	11°	−1.3	sltst.
T-56	355; 20+; 5.6	357; 17+; 5.0	2°	3°	−0.6	v.f.-f. ss.
	Average difference		4.9°	3.2°	−2.1	

that the angular difference in inclination is less than 180° may mean that some samples contain an unstable component of magnetization.

Stability during six-months interval.—Eighteen samples were measured after a six-months interval to determine if there had been any change in the direction and the intensity of magnetization. These measurements reveal (Table III) the lack of a temporary (Creer, 1959, p. 261) component of magnetization, except for one sample (T-41). Excluding sample T-41, the average differences in the two sets of measurements are 2.6° in declination, 2.9° in inclination, and −1.4 (×10⁻⁶ emu/cc) in intensity. The change in intensity of magnetization during the 6-month interval is the result of instrumental factors, except for sample T-41.

GENERAL TREATMENT OF CORRELATION DATA

Definition of normal and reverse magnetization.—In the Northern Hemisphere, "normal" rocks are those with N-seeking poles positively inclined downward (I, +ve), and "reversed" rocks are those with N-seeking poles negatively inclined upward (I, −ve). In this study the scatter of magnetization vectors is wide, and magnetic directions are somewhat arbitrarily defined. Normal magnetic directions are those with declinations between about 310° and 30°, and with inclinations in the lower hemisphere (+) of a stereonet (I, 0° to +90°). A few magnetic directions with shallow inclinations in the upper hemisphere and similar declinations (310°−30°) have been classed as normal. The best mean direction determined statistically from the gross data for normal magnetic directions so defined is 348° declination and +39° inclination.

"Reversed" magnetic directions are defined as those with declinations between about 90° and 190°. The inclinations of these magnetic directions are predominantly negative, but some have shallow positive inclinations. A mean direction of about 141½° declination and −31° inclination represents the "best" approximation of reversed magnetic directions from the gross data.

Magnetic directions that are outside the arbitrary limits of the normal and reverse directions are called intermediate reverse or intermediate normal depending on which of the two dominant directions the points are nearest. Those magnetic directions that are about equidistant from normal and reverse are called intermediate. A few points within several normal or reversed paleomagnetic intervals are either intermediate normal or intermediate reverse. These isolated points were included in calculating the statistical mean direction of the paleomagnetic interval, and they are considered to be part of the particular paleomagnetic interval. It is only when the majority of the points in a significant interval of the stratigraphic succession are clearly neither normal nor reverse that the interval has been called intermediate normal or intermediate reverse.

General correlation technique.—The subdivision of the magnetic directions into "paleomagnetic

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

intervals," and their correlation from locality to locality, was of necessity somewhat subjective, and partly based on geologic considerations. The magnetic results are repeatable, and only extend and amplify the previous work of Collinson and Runcorn (1960) and Black (1960).

A brief summary of the steps taken to effect the correlations follows. Although the subdivision and correlation were in practice not a straightforward, orderly process, for brevity of explanation the steps are so listed.

1. Magnetic directions of all the samples from each section were plotted on separate stereonets. It was noted that there are two predominant directions of magnetization, and numerous intermediate directions, especially intermediate reverse directions. The normal, reverse, and intermediate directions, as previously defined, were established.

2. Magnetic directions for each locality were plotted on "strip-logs" (Fig. 3).

3. Boundaries between obvious normal and reverse orientations were marked. The intermediate magnetic directions were also separated on each column where intermediate directions are other than a local occurrence in an interval that is clearly either normal or reversed (Fig. 3).

4. It was noted that there is a normal magnetic interval above and below the Alcova Limestone Member in all of the stratigraphic sections. The persistence of this interval, the concentration of the magnetic directions, the proximity of this interval to the Alcova Limestone Member, and the rather constant thickness of the interval suggested that the interval could provisionally be considered to be about the same age from locality to locality.

5. The base of the normal magnetic interval noted in (4) was then used as a paleomagnetic datum, and the cross sections were plotted (Figs. 2, 4).

6. The boundaries between normal, reversed, and intermediate magnetic directions on each section were extended from locality to locality.

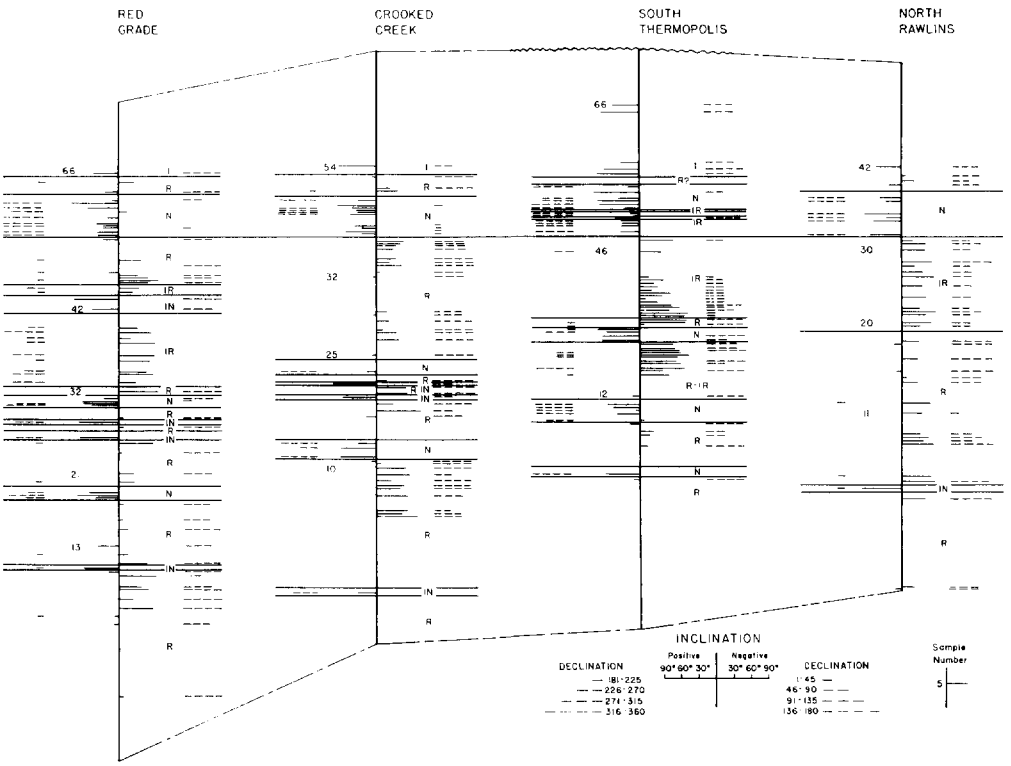


FIG. 3.—Cross section with "strip-logs" showing declination and inclination of each sample. Vertical scale and abbreviations same as on Figure 2. Datum is base of Interval 6. This cross section is basis for interpretation shown on Figure 2.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

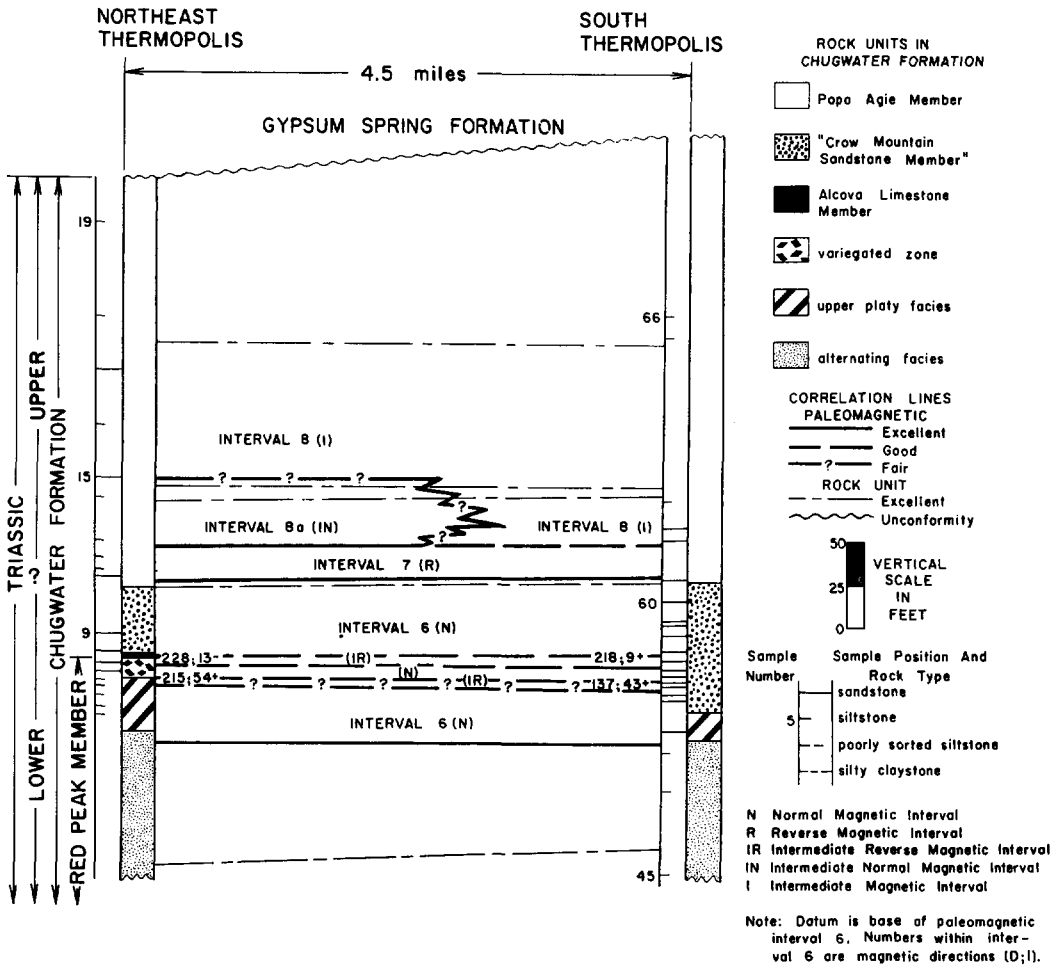


FIG. 4.—Cross section showing paleomagnetic and lithologic correlation between northeast and south Thermopolis.

The correlation procedure at this point was analogous to correlation by mechanical logs. It was assumed: (a) that paleomagnetic intervals do not cross each other; (b) that magnetic events are missing, or are not represented in the sampling, in some sections; and (c) that a conservative interpretation is desirable.

Method of presentation of data.—The magnetic orientation of each sample in a numbered interval on the cross sections is shown on a Schmidt stereonet (Figs. 5–16) for that interval. On some stereonets individual points are numbered; these numbers refer to the appropriate sample shown on the cross sections (Figs. 2, 4).

Statistical analysis of data.—The average direction of magnetization for each paleomagnetic interval shown on the cross sections (Figs. 2, 4) and the significance of this average direction were determined (Table IV) by a statistical method developed by Fisher (1953). The "angle of confidence" ($\alpha 95$) is the semi-angle of a cone whose apex is at the geographical point in question and whose axis is the average direction of magnetization of the analyzed interval (Doell, 1956, p. 159). The confidence angle equivalent to a 95 per cent probability is used in this paper, which means that there is 1 chance in 20 that the average (mean) direction of magnetization of a particular interval is outside the "cone of confidence" specified by $\alpha 95$ and the mean direction.

The number of samples (N) used in computing the mean direction of magnetization of an interval is shown on Table IV; each sample is from a different stratigraphic position. In the Fisher sta-

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

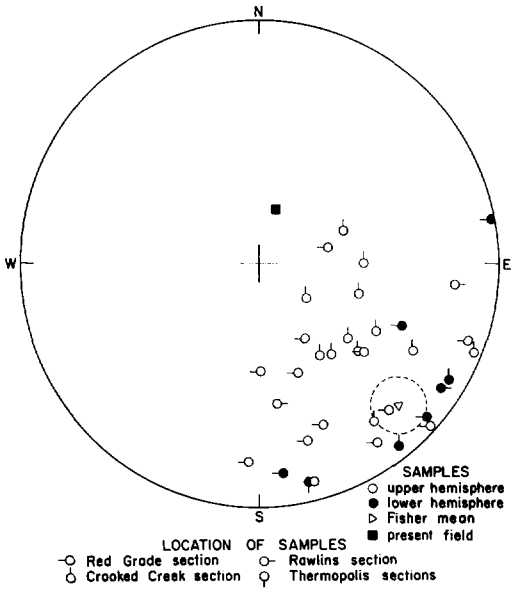


FIG. 5.—Directions of magnetization in interval 1. Symbols showing location of samples and present earth's field are the same on subsequent figures.

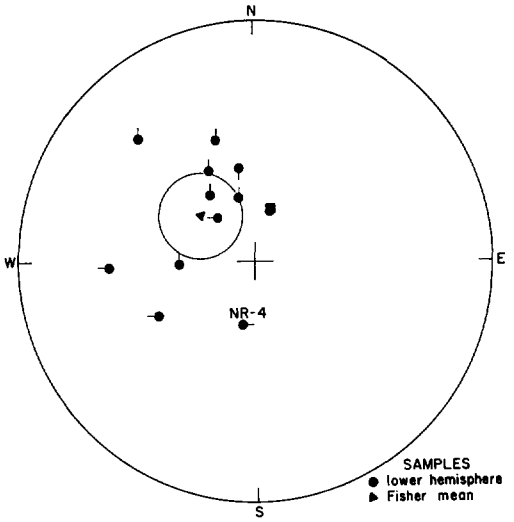


FIG. 6.—Directions of magnetization interval 2.

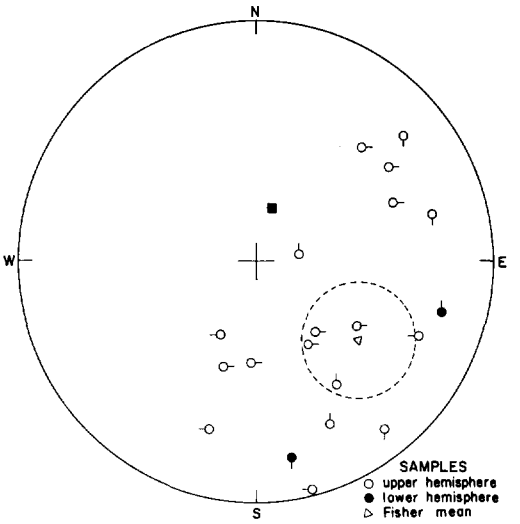


FIG. 7.—Directions of magnetization in interval 3.

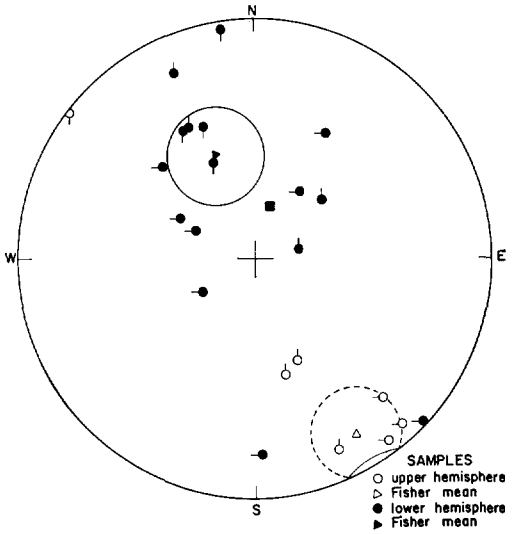


FIG. 8.—Directions of magnetization in interval 4.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

TABLE IV. AVERAGE DIRECTIONS OF MAGNETIZATION AND "ANGLE OF CONFIDENCE"

Paleomagnetic Interval Numbers	*Decl.*	*Incl.*	$\alpha 95$	k	N
8	$120\frac{1}{2}$	$+66$	13	13.5	11
7	160	$-17\frac{1}{2}$	$20\frac{1}{2}$	7.3	9
6	348	$+39$	7	10.4	47
5a	$253\frac{1}{2}$	-55	12	10.1	17
5	$145\frac{1}{2}$	-32	10	5.8	44
5c	$76\frac{1}{2}$	$-45\frac{1}{2}$	$8\frac{1}{2}$	10.7	29
5, 5a, 5c	$131\frac{1}{2}$	$-54\frac{1}{2}$	10	3.2	90
4 (Normal)	339	$+51\frac{1}{2}$	$17\frac{1}{2}$	5.7	15
4 (Reverse)	150	-16	$20\frac{1}{2}$	8.4	8
3	$127\frac{1}{2}$	$-44\frac{1}{2}$	22	3.2	20
2	310	$+65\frac{1}{2}$	15	9.2	12
1	$134\frac{1}{2}$	$-19\frac{1}{2}$	12	5.2	35
7, 5a, 5, 5c, 4 (Reverse), 3, 1	136	-43	6	3.4	162
6, 4 (Normal), 2	343	$+46$	$6\frac{1}{2}$	7.6	74
7, 5, 4 (Reverse), 3, 1	$141\frac{1}{2}$	-31	6	4.9	116

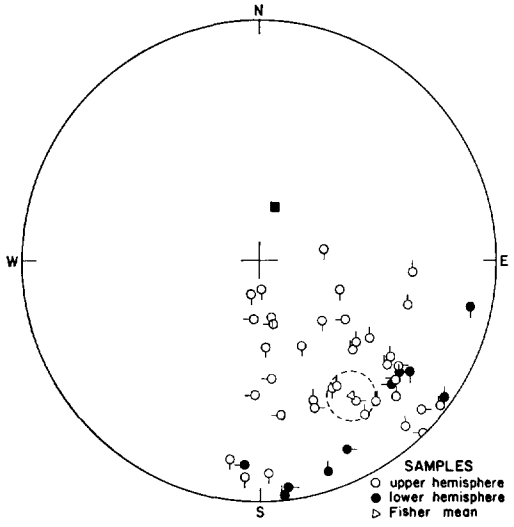


FIG. 9.—Directions of magnetization in interval 5.

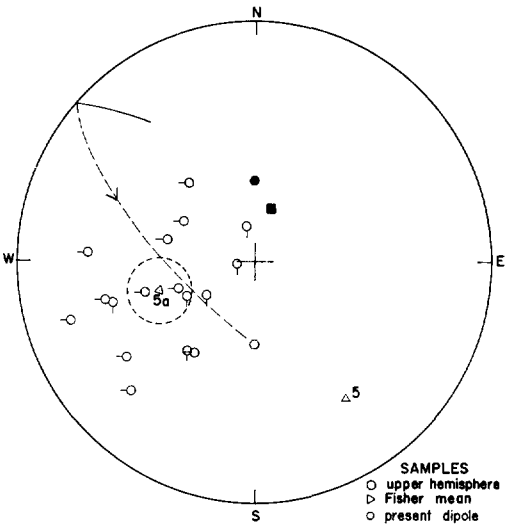


FIG. 11.—Directions of magnetization in interval 5a.

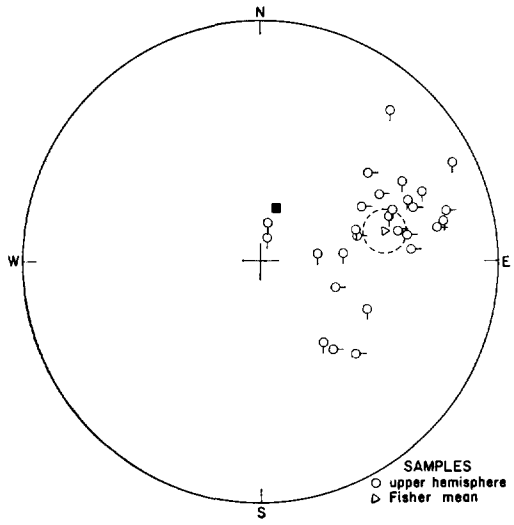


FIG. 10.—Directions of magnetization in interval 5c.

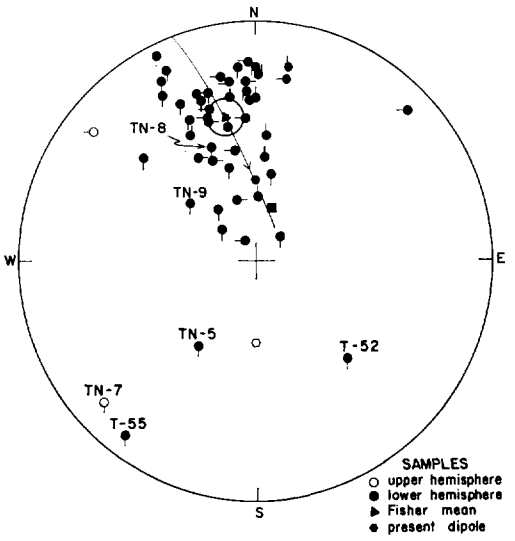


FIG. 12.—Directions of magnetization in interval 6.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

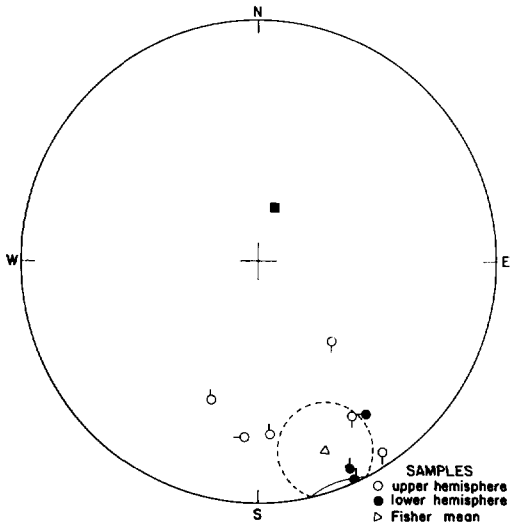


FIG. 13.—Directions of magnetization in interval 7.

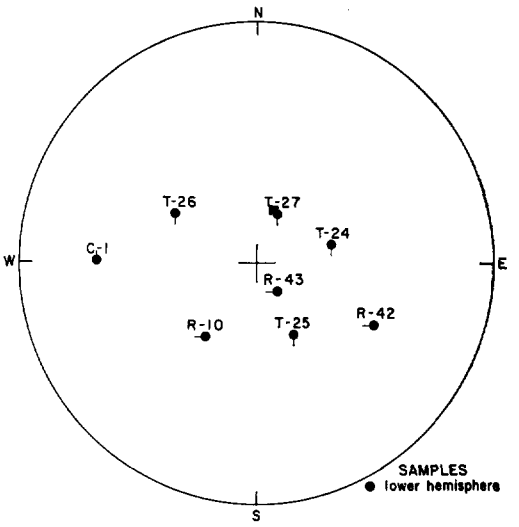


FIG. 15.—Directions of magnetization in intervals 1a, 1b, 5b, and 5d.

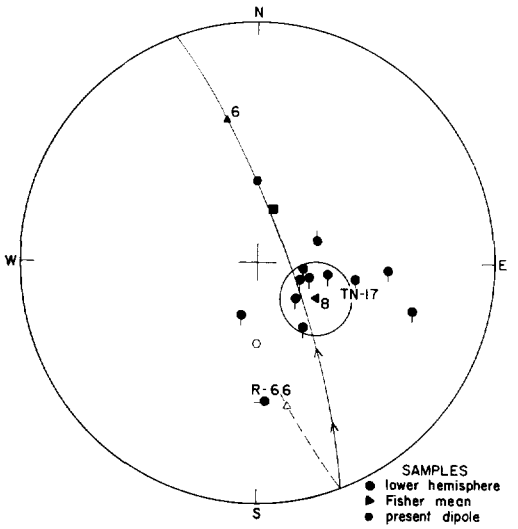


FIG. 14.—Directions of magnetization in interval 8.

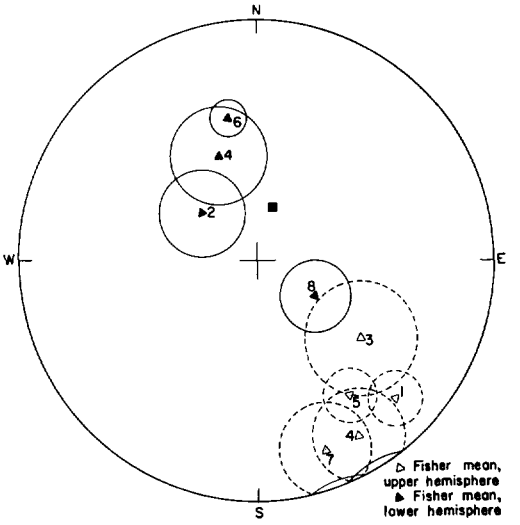


FIG. 16.—Summary of average directions of magnetization of all intervals except 5a and 5c.

tistical treatment, K is a constant called the precision parameter which describes the relative concentration of the magnetic directions about the mean direction. High values of K indicate high concentration; K=0 corresponds with a population uniformly distributed over the surface of a unit sphere. The quantity k shown in Table IV is the best estimate of the precision parameter.

LOCAL CORRELATION

General.—Correlation lines on key beds and the boundaries between paleomagnetic intervals are parallel. Both types of correlation lines are believed to be local "time-lines" between the northeast and south Thermopolis sections which are only 4.5 miles apart (Fig. 4).

Twenty-five feet of truncation at the top of

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

the Popo Agie Member is indicated between south and northeast Thermopolis, and the Gypsum Spring (Jurassic) Formation unconformably overlies the Popo Agie Member. Because of the truncation, the contact between the Popo Agie Member and the overlying Gypsum Spring Formation is not parallel with other correlation lines.

The Crow Mountain Sandstone Member is thicker at south Thermopolis than at northeast Thermopolis, and part of the Crow Mountain is a lateral facies equivalent of the Alcova Limestone Member of the northeast Thermopolis section. The Alcova Limestone Member is locally absent at south Thermopolis.

The absence of the variegated zone at south Thermopolis is believed to be related to the absence of the Alcova Limestone Member at south Thermopolis. The variegated zone is partly a secondary feature, formed by removal and remobilization of red pigment by solutions associated with the Alcova sea.

Details of paleomagnetic correlation.—The two thin intermediate reverse zones in each section within interval 6 (Fig. 4) are considered to be uncertain, because only one sample is reversely magnetized in each zone of the two sections. The two zones are included in the interpretation, however, because of the coincidence of the changes in magnetic direction in two sections, 4.5 miles apart.

The magnetic directions of the two samples (TN-13, TN-14) in interval 8a in the northeast Thermopolis section are closer to the mean direction for normally magnetized samples than they are to the mean direction for interval 8. The separation of interval 8a from the rest of interval 8 at northeast Thermopolis does not mean, however, that rocks the same age as those of interval 8a are not present at south Thermopolis. The separation is based on differences in magnetic directions.

In the south Thermopolis section rocks representing interval 7 were not sampled, and interval 7 has been extended into the south Thermopolis section from the northeast Thermopolis section. This is justifiable, because interval 7 is definitely present in the Red Grade, Crooked Creek, and northeast Thermopolis sections.

Summary.—(1) The paleomagnetic correlation is not as precise as the lithologic correlation. More lithologic horizons can be correlated between the two sections than are shown on the cross section. (2) The two types of correlation support each other. (3) Both types of correlation lines are approximately local "time-lines." (4) Additional sampling is needed to resolve several points regarding the paleomagnetic correlation. In parts of both sections it is not possible to obtain suitable samples for paleomagnetic measurements.

TABLE V. DIFFERENCES IN AVERAGE DECLINATION AND INCLINATION BETWEEN PALEOMAGNETIC INTERVALS

Intervals Compared	*Diff. in Decl.*	*Diff. in Incl.*
Interval 8—Interval 7	39½°	83½°
Interval 8—Interval 6	227½°	75°
Interval 7—Interval 6	188°	158½°
Interval 6—Interval 5 (excluding 5a, b, c, d)	202½°	173°
Interval 5a—Interval 5 (excluding 5b, c, d)	108°	23°
Interval 5c—Interval 5 (excluding 5a, b, d)	69°	13½°
Intervals 5, 5a, 5c—Interval 4 (normal)	207½°	177°
Interval 5 (excluding 5a, b, c, d)—Interval 4 (normal)	193½°	160½°
Interval 4 (normal)—Interval 4 (reverse)	189°	144½°
Interval 4 (normal)—Interval 3	211½°	173°
Interval 3—Interval 2	182½°	159°
Interval 2—Interval 1	175½°	134°

REGIONAL CORRELATION

General.—Details of each paleomagnetic interval not readily apparent on the stereonets of each interval (Figs. 5–15), or on the cross sections, are noted here. Some of the limits of the correlation are discussed. The differences in average (mean) declination and inclination between the paleomagnetic intervals are given in Table V. The differences in declination were measured clockwise; differences in inclination are the smallest angle between the average inclinations.

Interval 5.—Interval 5a (Figs. 2, 11) is separated from the rest of interval 5 because the average reversed magnetic direction in a 190-foot-thick sequence in the Red Grade section differs from the average reversed magnetic direction of interval 5 (excluding intervals 5b, c, d) by 108° in declination. The average inclinations of interval 5 and 5a are 23° apart. Similar magnetic directions were noted at south Thermopolis in four zones that are also assigned to interval 5a.

The directions of magnetization of the samples in interval 5a are along the great circle between

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin U

the mean direction of magnetization of interval 5 and the present dipole (Fig. 11). This interpretation implies that the samples of interval 5a were originally normally magnetized, and have migrated along the great circle toward the direction of reverse magnetization of the present dipole. The presence of the intermediate normally (?) magnetized interval 5b within interval 5a in the Red Grade section may support this interpretation. On the other hand, there is only one other normally magnetized interval (5d) in this part of the stratigraphic sequence in any of the sections. If interval 5a was originally a normally magnetized interval, most of interval 5a in the Red Grade section is probably not present in the other sections. The field evidence suggests, however, that rocks the same age as interval 5a at Red Grade are present in the other sections, although all of interval 5a at Red Grade may not be represented in sections that are northeast and southeast of the Red Grade section.

The anomalous magnetic directions assigned to interval 5c (Fig. 10) were found only in the south Thermopolis and north Rawlins sections (Fig. 2). Their vertical and lateral persistence is remarkable; their origin is not understood. Four possibilities for the origin of the magnetic directions of interval 5c are briefly discussed, although there are other possibilities. These are: (1) a unique position of the magnetic field; (2) a non-dipole component; (3) "streaking"; and (4) a detrital effect.

1. The magnetic directions of interval 5c are not believed to be the result of a unique position of the earth's magnetic field when the rocks acquired their original magnetization. "Normal" reversely magnetized directions are found in the Crooked Creek section in rocks that are believed to be the same age as interval 5c. The upper platy facies in the Red Grade section, which is probably equivalent to the upper part of interval 5c, also has "normal" reversely magnetized directions. Several scattered samples from all the stratigraphic sections have similar magnetic directions to interval 5c, a result that would be unexpected if the earth's magnetic field was only different during the time of magnetization of the rocks of interval 5c. These similar scattered samples show no consistent position with respect to magnetic intervals or to stratigraphy.

2. The possibility that the magnetic directions of interval 5c are the result of a non-dipole component during early Triassic time is rejected for the following reasons. The mean intensity of magnetization of red samples from interval 5c is 8.4×10^{-6} emu/cc, if four samples with intensities of magnetization greater than 25.0×10^{-6} emu/cc are ignored. The mean intensity of magnetization for all red samples from the Chugwater Formation is 6.5×10^{-6} emu/cc. The intensity of magnetization imparted by a non-dipole component should be generally less than the intensity of magnetization imparted by the dipole field. The localization of the effects of a non-dipole component to these two sections (north Rawlins; south Thermopolis) would also be difficult to visualize.

3. "Streaking," which is discussed later, would not appear to be an adequate explanation for these results. There is no apparent tendency for magnetic directions of samples from interval 5c to fall along a great circle between the earth's dipole and the direction of original magnetization.

4. The possibility that paleocurrents may have rotated detrital iron oxide minerals a consistent amount away from the dipole field is being investigated.

Samples from intervals 5a and 5c *may* contain an unstable magnetic component. As noted previously, other ways of accounting for these results are equally conjectural on the basis of the present data. It is of interest that the majority of the samples from intervals 5a and 5c have sublenticular-disturbed bedding. However, samples in interval 5 also commonly have this kind of bedding.

Interval 6.—The directions of magnetization of several samples suggest "streaking"; that is, the directions are along a great circle between the earth's dipole and the direction of original magnetization. "Streaking" indicates the possible presence of a secondary component induced by the present earth's field. Figure 12 shows a segment of the great circle between the mean direction of magnetization of interval 6 and the present dipole. The mean direction of magnetization of interval 6 is believed to more nearly approximate the magnetic field direction during Chugwater time than the other mean directions of magnetization. The magnetic directions of several samples in other paleomagnetic intervals (especially intervals 2 and 4; Figs. 6, 8) also suggest "streaking."

Interval 8.—The magnetic directions of samples in interval 8 are intermediate in inclination between normally and reversely magnetized samples

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

(Fig. 14). The magnetic directions are, however, along the great circle between the mean direction of magnetization of interval 6 and the present dipole (Fig. 14). Samples of interval 8 may have originally been reversely magnetized; subsequently, they may have migrated along the great circle toward the direction of normal magnetization of the present dipole. The intermediate position between normal and reversed magnetic directions of samples in interval 8, and the stratigraphic position of interval 8 above the reversely magnetized interval 7 (Fig. 13) support this interpretation. The mean intensity of magnetization of all red samples in interval 8 is 3.7×10^{-6} emu/cc, which is considerably less than the mean intensity of magnetization (6.5×10^{-6} emu/cc) of all the red samples collected from the Chugwater Formation. The range in intensity of magnetization in interval 8 is 0.6 to 6.6 ($\times10^{-6}$ emu/cc) for red samples.

Figure 16 summarizes the statistical data for each paleomagnetic interval.

Summary of paleomagnetic events.—The pertinent paleomagnetic data outlined in the preceding sections and shown on the cross sections (Figs. 2–4) can be summarized as follows. (1) Twenty-four magnetic events were distinguished in the five measured stratigraphic sections. Twelve magnetic events appear to record reversal of the earth's magnetic field. The cause of the other twelve magnetic events is uncertain, and further work is needed before these events can be accepted as reversal of the earth's magnetic field. (2) Only 3 of the more reliable magnetic events were found in the north Rawlins section, in contrast to 10 to 12 in each of the other sampled sections. This change can be explained by the larger sample-interval at north Rawlins than at the other three sections, and by thinning of correlative rock units toward the north Rawlins section. (3) Lateral or vertical changes in rock type do not affect the magnetic directions within some intervals. For example, the samples in interval 6 are red, sandy siltstone at Red Grade, red sandy siltstone and silty sandstone at Crooked Creek, and red, silty sandstone at south Thermopolis and at north Rawlins (Fig. 2). Whether changes in rock type play a role in the variations in magnetic directions in interval 5 is not known. (4) Magnetic events have different durations. Generally, periods of reversal of the earth's magnetic field appear to have been longer than normal periods. (5) Numerous magnetic events are probably not represented in the sampling. (6) The number of magnetic events that can be correlated seems to decrease as the distance between stratigraphic sections increases. A magnetic event may be represented in a section, but was not detected in the sampling, or a magnetic event may not be represented in a section because of unknown complexities of the sedimentation and the magnetization. (7) The average intensity of magnetization of 85 reversely magnetized, red siltstones is 8.9×10^{-6} (emu/cc). If the 6 most intensely magnetized samples (intensity greater than 25.0×10^{-6} emu/cc) are disregarded, the average intensity is 6.6×10^{-6} emu/cc. For comparison, the average intensity of magnetization of 33 normally magnetized, red siltstones is 6.8×10^{-6} emu/cc. The highest intensity of the normally magnetized, red siltstones is 20.5×10^{-6} emu/cc. Thus, intensity of magnetization appears to be the same in both normally and reversely magnetized samples (Fig. 17). (8) Figure 18 is a histogram of the intensities of magnetization of all the red samples collected from the Chugwater Formation. The mean intensity of magnetization is 6.5×10^{-6} emu/cc, if 13 samples with intensities of magnetization greater than 25.0×10^{-6} emu/cc are ignored. The standard deviation for this group is 3.4. For all red samples, the mean intensity of magnetization is 8.5×10^{-6} emu/cc.

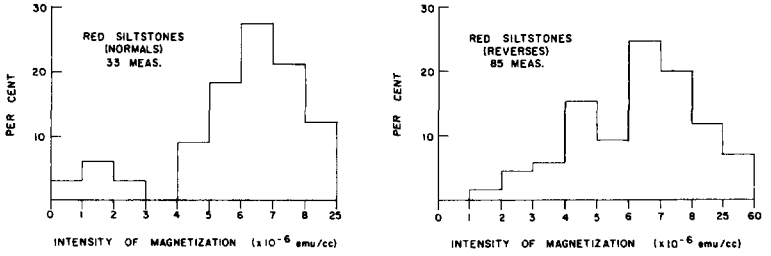


FIG. 17.—Histogram of intensity of magnetization in red, normally and reversely magnetized samples.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

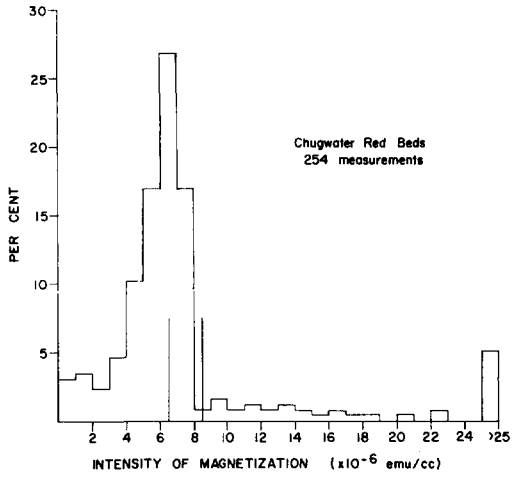


FIG. 18.—Histogram of intensity of magnetization of redbeds of Chugwater Formation. Mean intensities of magnetization shown by vertical lines.

STRATIGRAPHIC IMPLICATIONS OF REGIONAL PALEOMAGNETIC CORRELATION

Contemporaneity of paleomagnetic lines.—If the magnetization of a sedimentary rock is acquired at the time of deposition, or shortly thereafter, the boundaries between paleomagnetic events are synchronous. The paleomagnetic correlation between the south Thermopolis and the northeast Thermopolis sections (Fig. 4) indicates that the paleomagnetic lines are approximately synchronous. Evidence suggesting that the boundaries between the regional paleomagnetic intervals are also approximately synchronous is discussed in this section.

The general parallelism of the regional paleomagnetic lines with respect to each other, and the approximately equal thicknesses of rocks within a paleomagnetic interval in each section (Fig. 2) suggest that the over-all depositional rate may have been about the same throughout the area, and that the regional paleomagnetic lines may be approximately "time-lines." The non-parallelism of the boundaries of the rock units in the Red Peak Member, with respect to each other and with respect to the paleomagnetic correlation lines, suggests that the rock units vary in age from place to place. The lack of pronounced unconformities, the occurrence of lateral facies changes, and the gradational character of the rocks of the Red Peak Member support this view.

The incursion of the Alcova sea onto the shelf area is believed to have been a geologically short event. If the Alcova Limestone Member is approximately the same age from place to place, and if deposition was essentially continuous, the history of magnetic events adjacent to the Alcova Limestone Member can be summarized as follows (Fig. 19). (1) The time of magnetization was about the same in the Crooked Creek, south Thermopolis, and northeast Thermopolis sections. (2) The time of magnetization of the same rock units was slightly later in the Red Grade and north Rawlins sections than in the Crooked Creek, south Thermopolis, and northeast Thermopolis sections. If, however, the paleomagnetic correlation lines adjacent to the Alcova Limestone Member are approximately time equivalents of each other (Fig. 2), an alternative interpretation would be: (1) the Alcova Limestone Member and the variegated zone are younger in the Crooked Creek and northeast Thermopolis sections than they are in the Red Grade section; and (2) the Alcova Limestone Member and the variegated zone are about the same age in the north Rawlins and Red Grade sections. Present geological and paleomagnetic data do not permit deciding between these alternatives. Nevertheless, both interpretations suggest that the paleomagnetic correlation lines can be used as approximately regional "time-lines."

Summary interpretation of age relationships of rock units.—If the correlation within the Chugwater Formation is correct (Figs. 2, 4), the writer interprets the age relations of the rock units as follows. (1) On the basis of evidence from younger units, the silty claystone facies apparently is transgressive on the underlying formations (Fig. 2). (2) The lower platy facies and the lower part of the alternating facies are younger toward the southeast (north Rawlins) than they are toward the northwest (Red Grade). (3) The upper part of the alternating facies in the Red Grade section may be older than the upper part of the alternating facies in the Crooked Creek, south Thermopolis, and north Rawlins sections. (4) The upper platy facies in the Red Grade and north Rawlins sections may be older than the upper platy facies in the Crooked Creek and south Thermopolis sections. (5) The Alcova Limestone Member and the variegated zone are probably younger in the Crooked Creek and northeast Thermopolis sections than they are in the Red Grade section, and

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

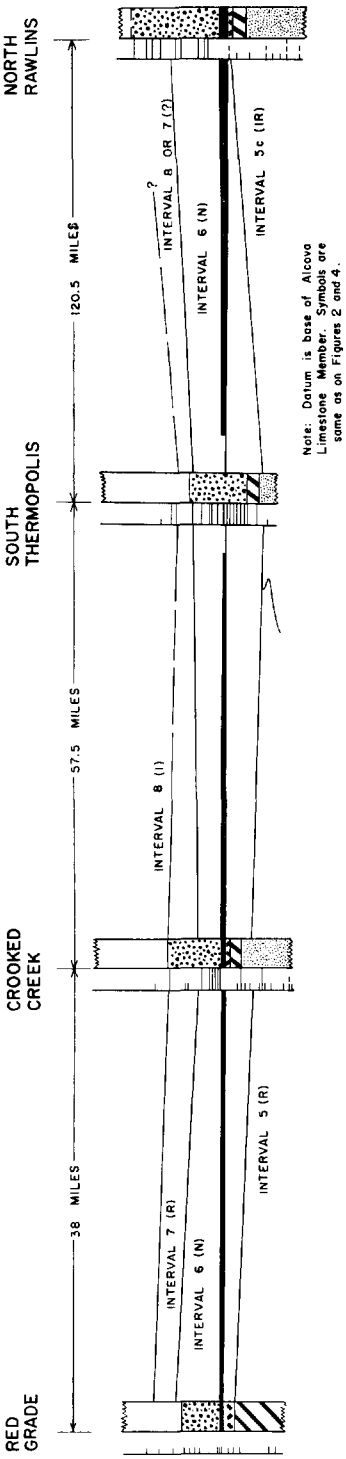


FIG. 19.—Regional cross section showing paleomagnetic intervals adjacent to Alcova Limestone Member, with base of Alcova Limestone Member as datum.

about the same age in the north Rawlins and Red Grade sections. (6) The Crow Mountain Sandstone Member probably differs in age from place to place (Figs. 2, 19), but the details of these changes are unknown. (7) A change in rock types near the boundary between intervals 7 and 8 suggests a regional change in the history of the Chugwater Formation. A change in magnetic directions also occurs above interval 7 (Fig. 14). The writer tentatively places the boundary between the Lower and Upper Triassic rocks at this position. Love (1957, correlation chart), with equal uncertainty, placed the boundary between the Lower and Upper Triassic rocks at the Alcova Limestone Member. Middle Triassic rocks are absent in Wyoming (McKee et al., 1959).

Investigation of Intervals Between Normal and Reversed Magnetic Directions

Momose (1958 et al., 1959) has found intermediate directions and intensities between normal and reversed magnetic directions of natural remanent magnetization in two sections of Pliocene volcanic rocks in central Japan. Similar results were previously reported by Sigurgeirsson (1957) from a study of Tertiary and Quaternary basaltic lava flows in Iceland. Intermediate directions of natural remanent magnetization have also been reported by Khramov (1958) from a study of sediments of western Turkmenia. In order to investigate the nature of the transition zone between normal and reversed magnetic directions in the Chugwater Formation, three intervals were sampled in some detail. These samples were taken at south Thermopolis, Red Grade, and Crooked Creek.

The results of this work are plotted on Figures 20–22, and can be summarized as follows. (1) Intermediate directions of magnetization occur in all three intervals, but the paths between normal and reversed magnetic directions traced by the intermediate directions appear to be different. (2) There is no evidence that the intensity of the samples in the intermediate zone is less than the intensity of normal and reversed samples (Figs. 20–22). Some intermediate samples are less intense, but others are equal to, or greater than, normal and reversed samples.

The change from normal to reverse magnetization, or *vice versa*, occurs over a stratigraphic interval of about 3.2 feet at Crooked Creek, 3.4 feet

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

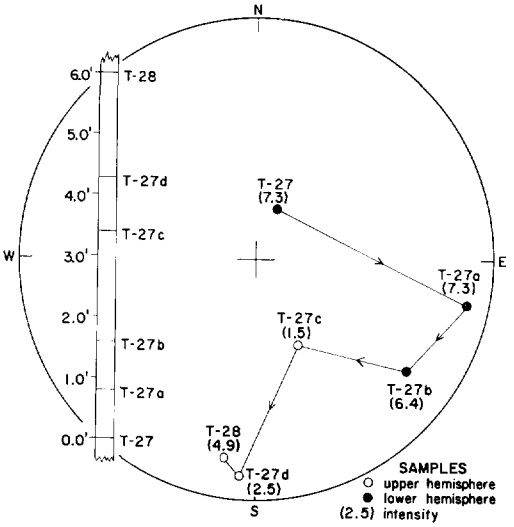


FIG. 20.—Directions of magnetization in transition zone from normal to reversed magnetization in south Thermopolis section.

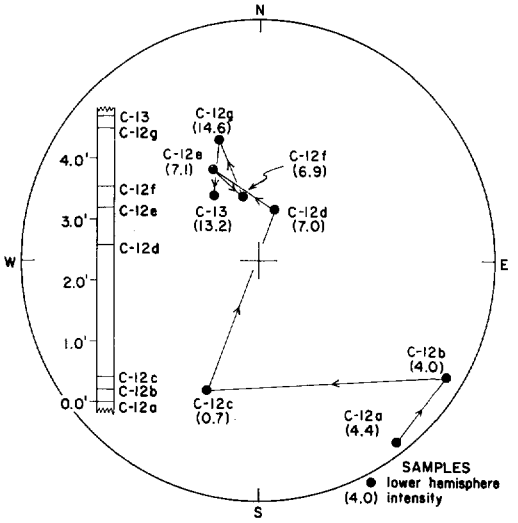


FIG. 21.—Directions of magnetization in transition zone from reversed to normal magnetization in Crooked Creek section.

at south Thermopolis, and 0.6 foot at Red Grade. The small stratigraphic interval at Red Grade possibly reflects different sedimentation rates and rock types, since several of the samples at Red Grade are silty claystone, which apparently accumulated more slowly than siltstone that occurs in the intermediate zone at south Thermopolis and Crooked Creek. However, if the dominant magnetization is chemical remanent magnetization (Nagata, 1961, p. 196–219; Cox and Doell, 1960, p. 652), the depositional rate was not a factor in the thickness of the intermediate interval between normal and reversed magnetic directions. Detailed measurements and stratigraphic positions of the samples are given in Appendix I.

In order to obtain an idea of the duration of the change from normal to reversed magnetic directions, the following data were used. (1) It was assumed that Lower Triassic rocks (excluding the Dinwoody Formation) were deposited in about 12 million years. This is 3 million years less than half of the time allotted by Kulp (1961, p. 1111) for Lower and Middle Triassic rocks. (2) It was assumed that the top of Lower Triassic rocks is about at the boundary between intervals 7 and 8. This placement of the upper boundary of Lower Triassic rocks gives an average thickness of 950 feet for Lower Triassic redbeds in this part of Wyoming. Applying these assumptions (12×10^6 /950 years/ft. of sedimentary rock), a depositional rate of 12,632 years for each foot of redbeds was obtained. This rate times the number of feet of redbeds in each transitional interval gives the following durations for the change from normal to reversed magnetic directions: $\simeq0.7\times10^4$ years (Red Grade); $\simeq4.0\times10^4$ years (Crooked Creek); and $\simeq4.3\times10^4$ years (south Thermopolis). The

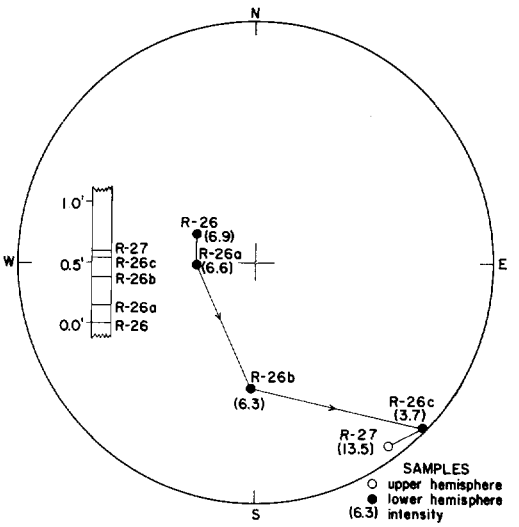


FIG. 22.—Directions of magnetization in transition zone from normal to reversed magnetization in Red Grade section.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

short duration at Red Grade was obtained without an adjustment for the possible effect (if any) of a slower rate of deposition of silty claystone as compared with siltstone. Although all the data and assumptions are approximate, the durations given may possess a modicum of heuristic value.

The duration of the Red Grade interval agrees with an estimate of 5,000–10,000 years calculated by Doell and Cox (1961, p. 286) for transitions between normal and reversed polarity of Icelandic basalts (Sigurgeirsson, 1957). Cox and Doell (1963, personal communication) now favor an upper estimate of the duration of the transitions for the Tertiary of about 2×10^4 years.

Intensity of Magnetization Compared with Rock and Bedding Types

Intensity and rock types.—The intensity of magnetization generally varies according to rock type. The median intensities ($\times10^{-6}$ emu/cc) from the cumulative curves (Fig. 23) of the various rock types are: drab (brown, yellow, and gray) siltstone and sandstone, 0.70; red, very fine-grained sandstone, 5.22; red siltstone, 6.66; and red, poorly sorted siltstone, 6.66. Ten silty claystones range in intensity from 5.4 to 70.4, but six samples are in the interval from 6.3 to 6.8. The number of silty claystone samples is too small to determine whether the median intensity of this group is larger or smaller than that of the red siltstones and poorly sorted siltstones. Since the range in intensities of the red siltstones and poorly-sorted siltstones is large, some samples are less intense than drab siltstone and sandstone, or red very fine-grained sandstone (Fig. 23). The variations in intensity according to rock type are, however, a good generalization.

Intensity and bedding types.—Horizontally laminated beds and "sublenticular-disturbed" beds appear to have about the same intensity of magnetization (Fig. 24). The median intensities ($\times10^{-6}$ emu/cc) are: "sublenticular-disturbed," 6.70; and horizontal-laminated, 7.30. The slightly higher intensity of magnetization in the horizontally laminated beds may mean that better sorting and packing of the grains in the horizontally laminated beds permitted a more uniform distribution and crystal ordering of hematite, with a consequent increase in intensity of magnetization. However, there may be no significant difference in intensity of magnetization between horizontal and "sublenticular-disturbed" beds. Apparently insufficient samples were studied.

The median intensity of magnetization of the cross-stratified, red siltstones and very fine-grained sandstones is 4.88. Since this value is close to the median intensity of red, very fine-grained sandstones (5.22), and since 74 per cent of the cross-stratified samples are this rock type, there seems to be no change in the intensity of magnetization because of cross-stratification.

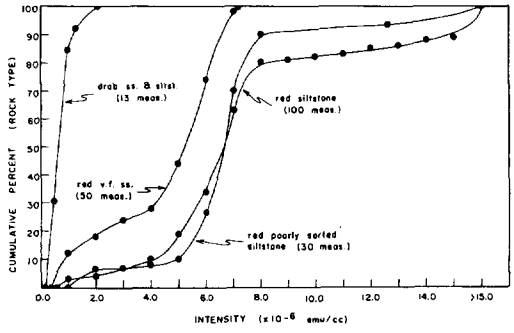


FIG. 23.—Cumulative curves of intensity of magnetization and rock type.

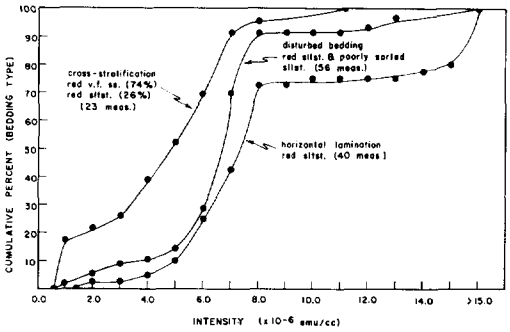


FIG. 24.—Cumulative curves of intensity of magnetization and bedding type.

Paleomagnetic Pole Positions

The results reported here probably do not give a precise Triassic pole position for west-central Wyoming. Many rocks that are useful for construction of a network for correlation purposes do not reliably indicate the position of the earth's magnetic field. Demagnetization experiments are necessary, and it may be desirable to exclude rocks that show instability.

With these limitations in mind, two pole positions have been calculated (Table VI) for comparison with other Triassic pole positions in the United States. The pole position derived from

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin U

interval 6 is considered to be the best of the two results, and it is closest to the pole positions of other workers. The variation in latitude of the poles from the North American Triassic is 22° and in longitude, 54°, if the longitude of the last entry in Table VI is ignored. Both of the pole positions calculated by the writer are farther east than the pole positions calculated by other workers.

Magnetization in Sedimentary Rocks

General.—Creer (1962, p. 86) has noted that although for paleomagnetic studies redbeds are the most useful of all sedimentary rocks, little is assuredly known about the origin of their natural remanent magnetization. The paucity of detailed data on the mineralogy of redbeds that are used for paleomagnetic studies accounts for much of the uncertainty about the origin of the magnetization. The magnetic properties of the individual iron oxides are better known than the mineralogy of specific redbeds, their diagenetic history, and the origin of their red pigment.

Depositional and chemical magnetization.—The two processes that will apparently yield a stable magnetization in sedimentary rocks are depositional remanent magnetization (DRM) and chemical remanent magnetization (CRM). Depositional (or detrital) remanent magnetization is the statistical alignment of ferromagnetic mineral grains along the direction of the earth's magnetic field during their deposition. The ferromagnetic mineral grains are assumed to be previously magnetized magnetic particles that attain a preferential alignment during deposition, and maintain this alignment after consolidation, giving the sedimentary rock a remanent magnetization (Cox and Doell, 1960, p. 652; Nagata, 1961, p. 222).

Relevant to chemical remanent magnetization, Martinez and Howell (1956, p. 205) and Doell (1956, p. 166) have shown that the magnetization of sedimentary rocks may be associated with chemical changes occurring after consolidation. Haigh (1958, p. 284–285) and Kobayashi (1959, p. 115–116) have indicated that a remanent magnetization is acquired by magnetic materials while undergoing a chemical change (e.g., reduction of hematite to magnetite) at constant temperature in a weak magnetic field.

The following explanation of the origin of the remanent magnetization of redbeds has been suggested by Creer (1961, p. 26–27). "It is suggested that the haematite grains at first were so small that they were superparamagnetic, many of them growing no larger than this. A small proportion of them grew large enough to pass through the viscous stage and then finally large enough to acquire a stable remanence in the direction of the then contemporaneous geomagnetic field. Part may be due to black detrital grains, which might possibly have been magnetized as the oxidation of the magnetite host proceeded."

More recently, Creer (1962, p. 99) expressed the belief that chemical magnetization can exist, but " . . . insufficient work has been done as yet to prove its existence in haematite or to estimate its importance in red beds in general." Creer also noted (1962, p. 99) a qualitative relationship between the amount of black, detrital ferromag-

Table VI. Pole Positions Compared with Other Triassic Paleomagnetic Poles from United States

Formation	Sampling Area		Pole Position				References
	Lat.	Long.	Lat.	Long.	δm	δp	
Moenkopi	Utah and Arizona		62N	103E	12½	12½	Cox and Doell (1960)
Connecticut Triassic	42N	73W	55N	88E	15	8	DuBois *et al.* (1957)
Upper Triassic lavas (Mass.)	42N	73W	55N	88E	11	6	Irving and Banks (1961)
Newark group (N.J.)	40½N	74.9W	63N	108E	4	3	Opdyke (1961)
Chugwater	Wyoming		48N	112½E	5	5	Collinson and Runcorn (1960)
Chugwater	43½N	109½W	51N	108E			Black (1960)
Chugwater (Interval 6)	43N	108½W	70N	142½E	8	5	This paper
Chugwater (Intervals 7, 5, 4 reverse, 3, 1)	43N	108½W	48N	171E	7	4	This paper

Notes: δm and δp are the errors at right angles to and in the direction of colatitude (Cox and Doell, 1960, p. 671). The pole for the Moenkopi Formation was calculated by Cox and Doell (1960) from data published in Collinson and Runcorn (1960), Kintzinger (1957), and Runcorn (1956). This table is modified from a table presented by Opdyke (1961, p. 1947).

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

TABLE VII. COMPARISON OF RESULTS FROM NON-DISTURBED AND "DISTURBED" BEDS

No.	*Bedding Type*	*Rock Type*	*Decl.; Incl.; Intensity* ($\times 10^{-6}$ *emu/cc*)	*Diff. in Decl.; Diff. in Incl.*	*Change in Intensity in "Dist." Unit*	*Remarks*
C-27	Wavy	sltst.	94; 35−; 17.2			C-28 1.0′
C-28	Disturbed	v.f. ss.	138; 8−; 7.0	44°; 27°	−10.2	above C-27
C-34	Horizontal	sltst.	133; 45−; 6.2			C-34 1.0′
C-35	Disturbed	p.s. sltst.	145; 23−; 7.3	12°; 22°	+1.1	below C-35
C-32a	"slightly disturbed"	p.s. sltst.	128; 26+; 6.4			From same
C-32d	Disturbed	p.s. sltst.	126; 4+; 5.4	2°; 22°	−1.0	bed
C-12a	Wavy	sltst.	142; 5+; 4.4			C-12a 2.5°
C-12b	Convolute	sltst.	121; 8+; 4.0	21°; 3°	−0.4	below C-12b
C-30	Horizontal	sltst.	174; 1+; 6.8			C-30 2.0′
C-31	Wavy	v.f.-f. ss.	134; 20−; 7.0	40°; 21°	+0.2	below C-31
C-3a	Horizontal	sltst.	131; 44−; 9.3			From same
C-3b	Convolute	sltst.	135; 50−; 6.9	4°; 6°	−2.4	bed
T-59	Horizontal	v.f. ss.	312; 36+; 3.6			T-59 1.8′
T-58	Cross-stratified	v.f. ss.	312; 74+; 1.1	0°; 38°	−2.5	above T-58
	Average difference			17.6°(D) 19.8°(I)	−2.17	

netic grains that are present, and the intensity of the natural remanent magnetization.

MAGNETIZATION OF CHUGWATER FORMATION

Generalization.—The writer proposes the following generalization for the origin of the consistent, measurable magnetization in the redbeds of the Chugwater Formation. The magnetization is predominantly a chemical remanent magnetization that was induced diagenetically, commencing soon after deposition. There is also a minor component of depositional remanent magnetization, and the magnetic directions measured are resultants of these two components. The evidence for this view is scant, and is as follows.

Magnetic results from non-disturbed and "disturbed" beds.—Pairs of samples from the same bed, or from closely adjacent beds, with the bedding more "disturbed" in one sample than in the other, were compared to determine if the magnetization was subsequent to the bedding disturbance. The results of this study are shown on Table VII. Some explanation, however, of the bedding types is needed. Sublenticular laminated is a term (Van Straaten, 1954, p. 34) for separate laminae of sand and mud, ranging in thickness from a few centimeters to fractions of a millimeter. Very little mud and sand occur in the massive siltstones of the alternating facies, and silt composes most of the rock. Disturbed bedding is characteristically found in the sublenticular laminated beds (Fig. 25). Bedding called disturbed and "slightly disturbed" in Table VII is "sublenticular-disturbed."

Descriptively, " . . . strata forming series of folds with nearly vertical axes are termed *convolute bedding*" (McKee, 1962, p. D152). The term, convolute bedding, as used in Table VII generally fits this definition, but the samples (Fig. 26) are more disturbed than the definition



FIG. 25.—Disturbed bedding and sublenticular laminations in massive siltstone of Red Peak Member. Maximum dimension of rock is 4.9 inches.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

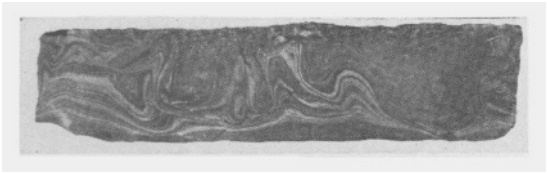


FIG. 26.—Convolute bedding in Red Peak Member. Same kind as is noted in Table VII. Maximum dimension of rock is 4.7 inches.

implies. Wavy bedding is a descriptive term indicating minor crinkling; it is intermediate between horizontal laminations and convolute bedding.

The average difference in magnetic directions of all of the pairs (17.6° in declination; 19.8° in inclination) is more consistent with the assumption that the dominant magnetization occurred after the bedding "disturbance" than before. The average differences in the two pairs collected from the same bed are: 3° in declination, 14° in inclination, and -1.7 ($\times 10^{-6}$ emu/cc) in intensity of magnetization. Although the data do not prove the assumption, it is difficult to conceive a primary depositional direction of magnetization that would not have been altered by the kind of bedding disturbance noted in some of the samples compared. Graham (1955, p. 338–339; 1956, p. 738) made similar observations on other rocks.

Although the lower values for the intensity of magnetization of five of the "disturbed" samples of five pairs (Table VII) may indicate a slight difference in ease of magnetization between non-disturbed and disturbed beds, probably no significance should be given to this result because of the small difference in intensity of magnetization.

Each sample of a pair was measured within a few minutes of one another, and several samples were measured more than once. The directions of magnetization in those samples measured more than once did not vary more than 2°; the intensities did not vary more than 3 per cent.

Other observations.—There appears to be no large difference in the intensity of magnetization between horizontally laminated and "sublenticular-disturbed" beds (Fig. 24). If the magnetization were predominantly depositional remanent magnetization, it should be greatly reduced with the destruction of the original depositional fabric of many of these rocks.

The intensity of magnetization is very low in several drab samples. Consequently, the magnetic directions are difficult to measure, and are greatly different from the magnetic directions in adjacent redbeds in some examples. Therefore, the introduction of the red pigment (whenever this occurred) may have been the initial step in the acquisition of a consistent, measurable magnetization.

There is a general indication that the intensity of magnetization increases with an increase in the amount of opaque minerals present (Table II). This may not mean, however, that the predominant magnetization is not chemical remanent magnetization. The three samples with the highest intensities of magnetization and highest content of opaque minerals are characterized by concentration of opaque minerals in thin laminae parallel with bedding. Most drab beds contain less opaque minerals than redbeds, but some drab beds, with barely measurable natural remanent magnetization, contain as great an amount of opaque minerals as some redbeds.

TIME OF MAGNETIZATION

The consistent magnetic directions in "disturbed" beds and in adjacent non-disturbed beds probably indicate that the dominant magnetization occurred later than the bedding disturbance. The regional correlation of numerous magnetic events suggests that the magnetization may have occurred moderately soon after deposition. The rate of chemical magnetization may be related to the rate of deposition, and chemical magnetization may generally parallel the sediment-water interface at an unknown depth below it.

If the dominant magnetization was depositional remanent magnetization, the time of magnetization was contemporaneous with the deposition, although the magnetic directions were modified during compaction (inclination reduced). Later chemical changes during diagenesis, when the earth's field may have been reversed with respect to an original depositional remanent magnetization, might impose either an intermediate magnetic direction, or completely "reverse" the original depositional remanent magnetization.

CONCLUSIONS

1. Paleomagnetic correlation lines in the local area between the northeast Thermopolis and the south Thermopolis sections are approximately "time-lines." The paleomagnetic correlation is not as precise as the lithologic correlation, but the two types of correlation do not conflict.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

2. Twenty-four magnetic events were found in the five measured stratigraphic sections of the Chugwater Formation. Twelve magnetic events appear to record reversal of the earth's magnetic field; the cause of the other twelve magnetic events is uncertain.

Other paleomagnetic conclusions from the regional study can be summarized as follows. (a) Lateral or vertical changes in rock types do not affect the magnetic directions within some intervals. (b) Magnetic events had different durations. (c) Numerous magnetic events are probably not represented in the sampling. (d) The number of magnetic events that can be correlated seems to decrease as the distance between stratigraphic sections increases. (e) The intensity of magnetization appears to be the same in both normally and reversely magnetized samples.

3. It is suggested that the paleomagnetic correlation lines can be used as approximately regional "time-lines." If this is valid, the age relations of the various units in the Chugwater Formation can be interpreted as follows. (a) The silty claystone facies, the lower platy facies, and the lower part of the alternating facies are younger toward the southeast (north Rawlins) than they are toward the northwest (Red Grade). (b) The upper part of the alternating facies may be about the same age in the Crooked Creek, south Thermopolis and north Rawlins sections, but probably older in the Red Grade section. (c) The upper platy facies in the Red Grade and north Rawlins sections may be older than the upper platy facies in the Crooked Creek and south Thermopolis sections. (d) The Alcova Limestone Member and the variegated zone are probably younger in the Crooked Creek and northeast Thermopolis sections than they are in the Red Grade section, and about the same age in the north Rawlins and Red Grade sections. (e) The Crow Mountain Sandstone Member apparently differs in age from place to place, but the details of these changes are not known. (f) The boundary between the Lower and Upper Triassic rocks has tentatively been placed near the top of the Crow Mountain Sandstone Member.

4. Intermediate directions of magnetization occur in zones between normal and reversed magnetic directions. The paths traced by these intermediate directions appear to be different. Some intermediate samples are less intense, but others are equal to, or greater in intensity than, normal and reversed samples.

5. Pairs of samples from the same bed, or from closely adjacent beds, in which the bedding of one of the samples is more "disturbed" than the bedding in the other sample, show average differences of 17.6° in declination, and 19.8° in inclination. The time of magnetization, which respect to the various kinds of bedding "disturbance," can not be determined solely from this data.

6. The intensity of magnetization generally varies according to rock type. From least intense to most intense, this variation is: (a) drab siltstone and sandstone; (b) red, very fine-grained sandstone; and (c) red siltstone and poorly sorted siltstone.

7. There may be no significant difference in intensity of magnetization between horizontal and "sublenticular-disturbed" beds. The median intensity of magnetization of cross-stratified, red siltstones and very fine-grained sandstones is less than the median intensity of magnetization of horizontal and "sublenticular-disturbed" beds, but this difference in intensity of magnetization is believed to be due to the predominance of very fine-grained sandstone in the cross-stratified samples.

8. Hematite is the dominant iron oxide in the red siltstones and very fine-grained sandstones. Ilmenite and magnetite also occur.

Appendix I

Results from Samples in Normal to Reverse or Reverse to Normal Detailed Intervals

The data for each sample are summarized as follows. An entry in the appendix contains (in order): sample number; stratigraphic position; declination; inclination; intensity ($\times 10^{-6}$ emu/cc); rock type; and the number of the interval on the cross sections to which the point is assigned.

C-12a; −458′ below base Alcova Limestone Member; 142; 5+; 4.4; sltst.; 1.
C-12b; 0.20′ above C-12a; 121; 8+; 4.0 sltst.; 1.
C-12c; 0.21′ above C-12b; 202; 41+; 0.7; sltst.
C-12d; 2.17′ above C-12c; 17; 72+; 7.0; sltst.
C-12e; 0.62′ above C-12d; 333; 55+; 7.1; p.s. sltst.; 2.
C-12f; 0.34′ above C-12e; 346; 68+; 6.9; sltst.; 2.
C-12g; 0.96′ above C-12f; 342; 46+; 14.6; sltst.; 2.
C-13; 0.20′ above C-12g; 325; 63+; 13.2; sltst.; 2.

T-27; −528′ below base Gypsum Spring Formation; 23; 72+; 7.3 sltst.; 5d.
T-27a; 0.80′ above T-27; 102; 10+; 7.3; sltst.
T-27b; 0.79′ above T-27a; 126; 22+; 6.4; sltst.
T-27c; 1.80′ above T-27b; 153; 57-; 1.5; sltst.; 5.
T-27d; 0.88′ above T-27c; 184; 10-; 2.5; sltst.; 5.
T-28; 1.71′ above T-27d; 189; 17-; 4.9; sltst.; 5.

R-26; −361′ below base Alcova Limestone Member; 296; 68+; 6.9; p.s. sltst; 4.
R-26a; 0.15′ above R-26; 268; 70+; 6.6; slty. clyst.; 4.
R-26b; 0.23′ above R-26a; 182; 46+; 6.3; slty. clyst.; 4.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin

R-26c; 0.16′ above R-26b; 134; 1+; 3.7; sltst. (mot. yel. gy.); 4.
R-27; 0.06′ above R-26c; 143; 5-; 13.5; sltst.; 4.

References

Black, J. S., 1960, Paleomagnetic study of the Triassic Chugwater Formation of Fremont County, Wyoming: Unpub. B.S. Thesis, Princeton Univ., Princeton, N. J., 76 p.

Collinson, D. W., and Runcorn, S. K., 1960, Polar wandering and continental drift: evidence from paleomagnetic observations in the United States: Geol. Soc. America Bull., v. 71, p. 915–958.

Cox, A., and Doell, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., p. 645–768.

Creer, K. M., 1959, A. C. demagnetization of unstable Triassic Keuper marls from S. W. England: Jour. Royal Astron. Soc., v. 2, no. 4, p. 261–275.

——— 1961, Superparamagnetism in red sandstones: Jour. Royal Astron. Soc., v. 5, no. 1, p. 16–28.

——— 1962, On the origin of the magnetization of red beds: Jour. Geomagnetism and Geoelectricity, v. 13, nos. 3, 4, p. 86–100.

Doell, R. R., 1956, Remanent magnetization of the upper-Miocene "blue" sandstones of California: Am. Geophys. Union Trans., v. 37, no. 2, p. 156–167.

——— and Cox, A., 1961, Paleomagnetism, *in* Landsberg, H. E., and Mieghem, J. Van, Advances in geophysics: New York and London, Acad. Press, v. 8, 392 p.

DuBois, P. M., Irving, E., Opdyke, N. D., Runcorn, S. K., and Banks, M. R., 1957, The geomagnetic field in Upper Triassic times in the United States: Nature, 180, p. 1186–1187.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. London Proc., ser. A, v. 217, p. 295–305.

Goddard, E. N., et al., 1948, Rock color chart: Geol. Soc. America.

Graham, J. W., 1949, The stability and significance of magnetism in sedimentary rocks: Jour. Geophys. Res., v. 54, p. 131–167.

——— 1955, Evidence of polar shift since Triassic time: Jour. Geophys. Res., v. 60, no. 3, p. 329–347.

——— 1956, Paleomagnetism and magnetostriction: Jour. Geophys. Res., v. 61, no. 4, p. 735–739.

Haigh, G., 1958, The process of magnetization by chemical change: Philos. Mag., v. 3, p. 267–286.

Irving, E., and Banks, M. R., 1961, Paleomagnetic results from the Upper Triassic lavas of Massachusetts: Jour. Geophys. Res., v. 66, no. 6, p. 1935–1939.

Khramov, A. N., 1958, Paleomagnetic correlation of sedimentary rocks: Leningrad, GOSTOPTEHIZDAT, 218 p. (English trans. by Australian Natl. Univ., 1960).

Kintzinger, P. R., 1957, Paleomagnetic survey of Triassic rocks from Arizona: Geol. Soc. America Bull., v. 68, p. 931–932.

Kobayashi, K., 1959, Chemical remanent magnetization of ferromagnetic minerals and its application to rock magnetism: Jour. Geomagnetism and Geoelectricity (Kyoto), v. 10, p. 99–117.

Kulp, J. L., 1961, Geologic time scale: Science, v. 133, no. 3459, p. 1105–1114.

Kummel, B., 1954, Triassic stratigraphy of southeastern Idaho and adjacent areas: U. S. Geol. Survey Prof. Paper 254-H, p. 165–194.

Love, J. D., 1939, Geology along the southern margin of the Absaroka Range, Wyoming: Geol. Soc. America Special Paper 20, 134 p.

——— 1957, Stratigraphy and correlation of Triassic rocks in central Wyoming, *in* Twelfth Ann. Field Conf. Guidebook: Wyo. Geol. Assoc., p. 39–45.

Martinez, J. D., and Howell, L. G., 1956, Palaeomagnetism of chemical sediments: Nature, v. 178, p. 204–205.

McKee, E. D., et al., 1959, Paleotectonic maps of the Triassic system: U. S. Geol. Survey Misc. Geol. Inv., Map I-300, 33 p., 8 pls.

McKee, E. D., Reynolds, M. A., and Baker, C. H., Jr., 1962, Laboratory studies on deformation in unconsolidated sediment: U. S. Geol. Survey Prof. Paper 450-D, p. D151–D155.

Momose, K., 1958, Paleomagnetic researches for the Pliocene volcanic rocks in central Japan: Jour. Geomagnetism, and Geoelectricity (Kyoto), v. 10, p. 12–19.

——— Kobayashi, K., and Yamada, T., 1959, Palaeomagnetic and geologic researches for the volcanic rocks around Lake Suwa—Palaeomagnetic researches for the Pliocene volcanic rocks in central Japan: Bull. Earthquake Res. Inst. (Tokyo), v. 37, p. 433–481.

Nagata, T., 1961, Rock magnetism: Tokyo, Maruzen Co. Ltd., 350 p.

Opdyke, N. D., 1961, The paleomagnetism of the New Jersey Triassic: a field study of the inclination error in red sediments: Jour. Geophys. Res., v. 66, no. 6, p. 1941–1949.

Picard, M. D., 1963, Paleomagnetic correlation of units within the Chugwater Formation, west-central Wyoming: Unpub. Ph.D. Thesis, Princeton Univ., Princeton, N. J., 117 p.

——— and High, L. R., 1963, Rhythmic alternation in the Triassic Chugwater and Brunswick formations, Wyoming and New Jersey: Contributions to Geology, v. 2, no. 1, p. 87–99.

Runcorn, S. K., 1956, Paleomagnetic survey in Arizona and Utah, preliminary results: Geol. Soc. America Bull., v. 67, p. 301–316.

Sigurgeirsson, Th., 1957, Directions of magnetization in Icelandic basalts: Advances in Physics, v. 6, p. 240–246.

Van Houten, F. B., 1961, Ferric oxides in red beds as paleomagnetic data: Jour. Sed. Petrology, p. 296–300.

Van Straaten, L. M. J. U., 1954, Sedimentology of Recent tidal flat deposits and the psammites du Condroz (Devonian): Geologie En Mijnbouw (NW. Ser.), p. 25–47.

Downloaded from http://pubs.geoscienceworld.org/aapg/aapgbull/article-pdf/48/3/269/4395435/aapg_1964_0048_0003_0269.pdf by Curtin